Text Analysis

Fundamentals of Computing and Data Display

10/9/2024

Text Data

- Usually extracted using APIs (e.g., NYT API) or web scraping (e.g., Wikipedia for Assignment 3).
- Many techniques require cleaning … but some more sophisticated techniques do not.
- Lots of different goals with text data – make sure you are using the appropriate one!

Topic Modeling

- **Idea**: We want to understand the topics within the documents of a large corpus of text. We could read them all … or we can use topic modeling to identify groups of documents based on most frequent words.

- A form of **unsupervised machine learning**, similar to **clustering**.

Traditional Sentiment Analysis

- Dictionary-based, with words containing a certain sentiment.
- Accounts for negation.
- Can suffer from changing ways in which words are used.
- Might not work as well with sarcasm or when in reference to other things (like images).

Slang and Context

- Methods discussed won’t really catch slang or context very well.
- Bigrams and n-grams can help this a little bit
  - “supreme court” or “commander in chief”
- Usually, individual units, or tokens, are individual words, but this doesn’t have to be the case
  - May also include emojis as individual tokens

Limitations

- Traditional methods are a bit limited … But they can still be useful!
- Topic modeling won’t tell you the exact topics, but they can give you a good idea of what is going in within the whole corpus.
- Just cleaning text and looking at frequent words can also be extremely useful.

More Sophisticated Methods

- Neural networks such as transformers can capture more of the context and meaning within text, and these can provide more sophisticated forms of doing things like sentiment analysis.
- Benefits of model-based approach to sentiment analysis
  - Can be built with current data to better reflect modern sayings
  - More flexible in assigning sentiment
  - Includes more context rather than word by word

- **Note**: this does not require the same preprocessing!

Resources

- For handling text data: https://www.tidytextmining.com
- Hugging Face: https://huggingface.co